END SEMESTER EXAMINATION May-June – 2025

Name of Program: B Tech (Biotechnology) Semester: 4

Name of Paper: Programming for computational Biology Paper Code: TBT408b

Time: 3 Hrs] [Maximum Marks: 100

Note:

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

1.	20 Marks	CO-1
(a)	Define an interpreter. Discuss Python's approach to indentation and how it differs from other programming languages.	
(b)	What to you mean by keywords? Explain the role of keywords in Python programming and provide examples.	
(c)	Define the terms – identifiers, unary operators, binary operators, and constants.	
2.	20 Marks	CO-2

(a)	Explain the use of scope in a program. What is the value of x printed in the given program segment?
	`def my_function(x):`
	`  x = x+10`
	`  return x`
	
	`x = 20`
	`x = my_function(x+5)`
	`print(x)`	

(b)	Define a stack. Create a Python class `Stack` to implement a stack using a list. The class should support the following operations: push(item), pop(), peek(), is_empty(), `size()`, display().	
(c)	What is a user defined datatype? Explain with suitable example.	
3.	20 Marks	CO-3
(a)	Define the terms: list, tuple, set. Explain each with suitable program examples. Also compare these.	
(b)	Define a string in Python. Discuss any six methods of string data type with relevant examples.	
(c)	Write a Python program using function to count and print frequency Character in alphabetically ascending order.	

4.	20 Marks	CO-4
(a)	What is List Comprehension? Explain the concept - list Comprehension with suitable example of string.	
(b)	Define a tuple. Write a Python program to store 20 random integers in a list named NumsList. Then create two blank tuples named as EvenInts and OddInts. Copy the even numbers from the NumsList to EvenInts tuple and odd numbers to OddInts tuple.	
(c)	What is dictionary? Explain the basic concepts of dictionary with useful example.	
5.	20 Marks	CO-5
(a)	What is a file? Explain the basic methods of handling files in python.	
(b)	Write a Python program that reads the contents of a file and determines the frequency of each word. The program should process the text efficiently and display the word count in a structured manner.	
(c)	Define an Exception in Python. List any five Built-in Exceptions. Explain the NameError exception with simple example.	